Spencer Harman

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Education

North Carolina State University — Raleigh, NC

M.S. in Computer Science – Track in Security (ABM Program, GPA: 4.0) — *Jan. 2027 – Dec. 2027*

B.S. in Computer Science – Cybersecurity Concentration (GPA: 4.0) — *Aug. 2023 – Dec. 2026*

Experience

Information Security Analyst Intern | *Red Hat* | *Raleigh, NC* — *May 2026 – August 2026*

- **Cross-Functional Security:** Partnered with Red Hat/IBM Incident Response, Cyber Threat Intelligence, and Offensive Security teams to investigate active threats and streamline cross-departmental incident escalation.
- **SOAR & SIEM Automation:** Designed an automated credential-exposure detection system using Sumo Logic/Splunk and AWS, accelerating incident response time from **hours/days to seconds**.
- **Open-Source Security Tools:** Contributed core features to open-source tools (**LeakTK**) - **grep** search engine for ad-hoc pattern matching, **redaction** capabilities for exposed secrets, **login** authentication for pattern servers.
- **AppSec & AI Pentesting:** Executed targeted penetration testing on enterprise web applications and LLM/AI models, identifying critical vulnerabilities and delivering actionable remediation blueprints.
- **Malware Mitigation:** Conducted a cryptanalysis audit of Miasma/Shai-Hulud malware; deployed YARA rules and a supply chain security framework with CSIRT to block malicious packages across **thousands of endpoints**; Expanded **Indicator of Compromise** detection capacity and investigation thoroughness.

Software Engineer | *Liquid Rocketry Lab* | *Raleigh, NC* — *September 2024 – Present*

- **Firmware Development:** Developed and debugged bare-metal firmware for a custom microcontroller, optimizing real-time hardware control and data logging for the critical rocket test stand infrastructure.
- **RF & Telemetry:** Implemented secure wireless communications to bridge data streams between the mission control room and the rocket test stand, ensuring low-latency command execution.
- **Legacy Migration:** Architected backwards-compatibility layers within the storage file system to ensure seamless data parsing and integration across legacy firmware configurations.
- **Flight Computer R&D:** Spearheaded research and development for next-generation flight computers, focusing on sensor fusion, fault tolerance, and telemetry safety margins.

Technical Sales Intern | *Red Hat* | *Raleigh, NC* — *May 2025 – August 2025*

Projects

Web-to-Android Policy Violation Security Research | *Graduate Research Project, WSPR @ NC State*

- Investigating whether data supplied by JavaScript in Android WebViews crosses into native Android code and reaches privacy-sensitive app functionality.
- Designing an extension to WebViewTracer that combines static APK analysis with VisibleV8 and Frida runtime instrumentation to trace web-to-Android communication paths.
- Built a static analysis pipeline using YARA rules and JADX to identify WebView-related APIs and prioritize Android applications for dynamic analysis.

Phidippus Web Vulnerability Scanner | *Python, Asyncio, HTTPX, BeautifulSoup4, NIST CVE API*

- Developed an asynchronous web vulnerability scanner in Python, achieving **2,600+ public downloads.**
- Engineered scanning modules to detect critical OWASP Top 10 exploits, including error/boolean/time-based SQL injections, Cross-Site Scripting (XSS), cryptographic failures, and sensitive file path exposures.
- For scans requiring login credentials, an `httpx.AsynchClient` is created to preserve session cookies.
- Uses the NIST CVE API to relate findings to CVE reports and returns them in the final report.

Technical Skills

Languages: Python, Java, C, JavaScript, Swift, Assembly (x86/ARM), R, Bash, PHP, Machine Code, SQL/NoSQL
Technical Skills: Burp Suite, Kali Linux, Splunk, Wireshark, Network Security, Penetration Testing, Incident Response, Packet Analysis, Operating Systems, Reverse Engineering, Software Engineering, GDB, Valgrind, TCP/IP Protocol Suite, Packet Loss & Retransmission Analysis, Network Forensics, Socket Programming
Interpersonal Skills: Analytical Thinking, Problem Solving, Accountability, Communication, Coachable
Security Community & CTF: ISACA Member (Sep. 2024 - Present), Capture The Flag Challenges (2nd place April 2025 HackPack CTF, HackTheBox, picoCTF)